Global Equity Market Co-movement During Crisis Periods

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Abstract

We analyze global equity market co-movement during the last 25 years using a dynamic spatial model. Based on a generalized autoregressive score model, we analyze the co-movement among global, European, American, and Asian equity markets during various crises including the Asian, the financial, the European sovereign debt crisis, as well as the economic turmoil due to the current Covid-19 pandemic. Our results show an increase in the co-movement prior to the onset of the global financial crisis in 2008, a generally higher co-movement among European countries compared to American and Asian-Pacific countries, while the highest level of co-movement is reached during the current pandemic. When accounting for time-varying variances, however, we find evidence for global contagion only during the Covid-19 pandemic which induces an economic downturn in the first quarter of 2020.

Keywords: Co-movement; Contagion; stock markets; crisis ; Volatility; Covid-19; GAS

JEL Classification: C21, G01, G15

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1 Introduction

Shortly after the first cases of unknown pneumonia were detected in the city of Wuhan, China, the pandemic induced by the Covid-19 virus was declared as a public health crisis of international concern. The impact on financial markets was unprecedented, causing simultaneous downturns within equity markets around the world.

Whether and when a higher level of co-movement between financial markets during crisis period gives evidence for contagion, has been widely discussed within financial market literature. Most prominently, Forbes and Rigobon (2002) define contagion as an increase in the correlation between two or more markets after a crisis event. They argue that the occurrence of a crisis prior to such an increase in interdependence, is what distinguishes contagion from an increase in co-movement due to deepened financial integration. They also note that contagion might falsely be detected due to a crisis induced increase of volatility within the considered markets. The subject of defining, measuring and testing for contagion has stirred numerous prominent contributions, including Bekaert *et al.* (2009), Bekaert *et al.* (2014), Bae *et al.* (2003), Longin and Solnik (2001). For a comprehensive survey we refer to Forbes (2012). The methodological approaches towards measuring co-movement and contagion are manifold, including linear correlations (e.g. Forbes and Rigobon, 2002), factor-models (e.g. Bekaert *et al.*, 2009, 2014), model-free or co-exceedance approaches (e.g. Bae *et al.*, 2003; Londono, 2019), Vector Autogregressive models (e.g. Boubaker *et al.*, 2016), and Copula (e.g. Rodriguez, 2007; Kenourgios *et al.*, 2011a). This paper uses a different approach. We model the complex global dependencies in international financial markets using spatial techniques. Spatial models are at the core of many recent studies in applied and theoretical econometric research (see, for instance, Kelejian and Prucha, 2010; Qu and Lee, 2015). These approaches have an extensive history in geography literature and since Krugman (1992b,a) seminal works, have gained attention in economics as well.

Applications to financial markets include Fernandez (2011), who tests spacial linkages between financial markets in Latin-America by relying on a spatial capital asset pricing model. Asgharian *et al.* (2013) applies a static spatial model to analyze the extent to which geological and economic relations between countries affect the co-movements among stock markets. The equity market is also studied by Arnold *et al.* (2013), who applies a spatial model within a risk management context. Recently, Blasques *et al.* (2016b) proposed an approach to model time-varying (spatial) dependence based on a generalized autoregressive score (GAS) framework. GAS models are observation driven models, which have been introduced by Creal *et al.* (2013) and Harvey (2013) and are becoming increasingly popular. Their merit lies in providing a general framework for time variation in parametric models, while offering straight forward estimation and inference, which renders them a suitable methodology for our analysis.

We complement the literature on equity market contagion by analyzing global time-varying co-movement among financial markets over multiple crisis periods between 1995 and 2020 using a time-varying spatial model. We examine global co-movement based on 38 stock market indices as well as subsets representative for the European, Asian-Pacific and American markets. Our contributions are i) to provide a global view by studying a large panel of stock indices over a long period of time including various crises, ii) applying a novel model, namely a time-varying spatial dependence model recently introduced by Blasques *et al.* (2016b), which allows us to analyze the evolution of co-movements among stock markets over time, and iii) assess the question of whether global contagion occurred during the major crises of the last 25 years. Implementing the time-varying spatial dependence model augmented by dynamic variances, we also account for increased volatility during crisis periods and calculate confidence bounds for the co-movement parameters based on Blasques *et al.* (2016a).

Our results reveal a general increase in co-movement during the last 25 years which is most pronounced for the European countries. Our time-varying approach shows that co-movement is time-varying also during crisis period often exhibiting several peaks within a crisis period. Allowing for time-varying variances within our spatial model, we also confirm the findings by Forbes and Rigobon (2002): We observe a widening of confidence bounds during events or crisis periods, indicating an increase in variances. This heteroscedasticity might bias contagion tests when not accounted for. When testing for contagion, we find evidence for global contagion during the Covid-19 pandemic, while for most of the remaining events, we conclude that we observe interdependence rather than contagion.

The remainder of this paper is structured as follows. Section 2 presents the methodology, Section 2.2 describes the data and preliminary analysis, Section 3 provides and discusses the results, while Section 4 concludes.

2 Spatial modeling of stock market co-movements

2.1 Modeling time-varying spatial dependence

Spatial models offer an interesting alternative to common approaches when measuring interdependence, because they deliver a measure of co-movement that goes beyond pair-wise correlations. Thereby, they are able to account for the complex dependence structure underlying global equity markets. Our analysis is based on the work of Blasques *et al.* (2016b), who introduces a dynamic spatial model based on the GAS framework offered in Creal *et al.* (2013) and Harvey (2013). In detail, we consider the spatial autoregressive model (SAR) to analyze spatial interactions across units (Cliff and Ord, 1973, 1981; Ord, 1975). The SAR provides means to comprise the impact of the spatial lag of a cross-section on the corresponding units within that cross-section in a single scalar parameter. Denote the SAR in period $t = 1, \ldots, T$ as:

$$y_t = \rho W y_t + X_t \beta + \varepsilon_t, \qquad \varepsilon_t \sim p_\varepsilon(\varepsilon_t; \Sigma, \lambda), \tag{1}$$

where y_t is a $(n \times 1)$ vector of cross-sectional observations, ρ is the endogenous spatial dependence parameter, W is a $(n \times n)$ spatial weights matrix, X_t is a $(n \times k)$ matrix of exogenous regressors, β is an unknown $(k \times 1)$ vector of coefficients, and ε_t is a $(n \times 1)$ vector of disturbances with mean zero, variance-covariance matrix Σ, and multivariate density $p_\varepsilon(\varepsilon_t; \Sigma, \lambda)$ including λ to further specify the distribution of the disturbances. The vector $W y_t$ is commonly referred to as the spatial lag of the dependent variable y_t, where the weight matrix W entails information on the spatial distances across units. Note, that the data generating process can be written as:

$$y_t = (I - \rho W)^{-1} X_t \beta + (I - \rho W)^{-1} \varepsilon_t, \tag{2}$$

where I is an $(n \times n)$ identity matrix. Restricting the estimation space of the spatial dependence parameter such that $-1/w^+_{\min} < \rho < 1$, where $w^+_{\min}$ is the absolute value of the smallest eigenvalue of W, imposes positive definiteness on the variance-covariance matrix (e.g., Lee, 2004; Lesage and Pace, 2009). Following Debreu and Herstein (1953), the SAR model can be expressed as an infinite power series to demonstrate the impact of spillovers based on the order of neighbors:

$$y_t = X_t \beta + \rho W X_t \beta + \rho^2 W^2 X_t \beta + \ldots + \varepsilon_t + \rho W \varepsilon_t + \rho^2 W^2 \varepsilon_t + \ldots. \tag{3}$$

For instance, unit i may impact units $j \neq i$ through ε_{it} and $x'_{it}\beta_{it}$, where ε_{it}, x_{it}, and β_{it} indicate the i-th row of ε_t, X_t, and β_t, respectively. Spillovers from unit i on its (first-order) neighbors are affected by ρ and the distance to units j as determined by the spatial weight matrix W. In turn, transmitted shocks may induce feedback effects on the second order neighbors of unit i which includes unit i itself.

In Equation 1, the impact of the weighted dependent variable Wy_t on y_t is captured by the static parameter ρ. The dynamic version of the SAR model based on the observation-driven GAS framework endows both the spatial dependence parameter and the variance-covariance matrix with time-varying dynamics (Blasques *et al.*, 2016b):

$$y_t = \rho_t W y_t + X_t \beta + \varepsilon_t, \qquad \varepsilon_t \sim p_\varepsilon(\varepsilon_t; \Sigma(f_t^\sigma), \lambda), \tag{4}$$

where $\rho_t = h(f_t)$. Thus, the time-varying spatial dependence between units is obtained as a strictly monotonous transformation of the dynamic parameter f_t. We use $h(f_t) = \tanh(f_t)$ as a reasonable choice for the transformation to constrain $\rho_t \in (-1, 1)$. Further, the covariance matrix is given by $\Sigma(f_t^\sigma) = diag\left(\sigma_1^2(f_{1t}^\sigma), \ldots, \sigma_n^2(f_{nt}^\sigma)\right)$ with $\sigma_i^2(f_{it}^\sigma) = \exp(f_{it}^\sigma)$ for $i = 1, \ldots, n$.

The generalized autoregressive score (GAS) framework is employed to model the evolution of f_t and f_t^σ over time (Creal *et al.*, 2011, 2013; Harvey, 2013). More precisely, we assume that the autoregressive updating function of f_t follows a GAS(1,1)-type process:

$$f_{t+1} = \omega + A S_t \nabla_t + B f_t, \tag{5}$$

where ω, A, and B are scalar parameters, S_t is a scale matrix, and ∇_t is the score function to link the direction of the steepest ascent of the observation density at time t to f_t. Similarly, the updating function of f_t^σ is assumed to follow an n-dimensional GAS(1,1)-type process:

$$f_{t+1}^\sigma = \omega^\sigma + A^\sigma S^\sigma \nabla_t^\sigma + B^\sigma f_t^\sigma, \tag{6}$$

where w^σ is a $(n \times 1)$ vector containing unit-specific intercepts, A^σ and B^σ are scalar parameters, S^σ is a scale matrix, and ∇_t^σ is the $(n\times1)$ score vector. Whilst intercepts are employed individually for each unit, joint A^σ and B^σ ensure parsimony of the adaptive model.

Both in Equation 5 and 6, we consider S_t and S^σ to be identity matrices. The (scaled) scores can then be denoted as the partial derivative of the log-likelihood function at time t with respect to f_t and f_t^σ, respectively:

$$\nabla_t = \frac{\partial \ell_t}{\partial \rho_t} \cdot \frac{\partial \rho_t}{\partial f_t}, \quad \nabla_t^\sigma = \frac{\partial \ell_t}{\partial f_t^\sigma}, \tag{7}$$

where $\partial\rho_t/\partial f_t = 1 - \tanh(f_t)^2$. Lastly, ℓ_t is given by:

$$\ell_t = \ln p_\varepsilon(\varepsilon_t; \Sigma(f_t^\sigma), \lambda) + \ln |(I - \rho_t W)|. \tag{8}$$

Considering the distribution of the disturbance vector, ε_t, we assume a multivariate t-distribution with ν degrees of freedom, where ν is estimated along with the other model parameters. This distributional choice seems appropriate, since the stock return series in our application exhibit excess kurtosis (Harvey and Luati, 2014). We estimate the unknown parameter vector $\theta = (\beta, \omega, A, B, \omega^\sigma, A^\sigma, B^\sigma; \lambda)$ using Maximum Likelihood estimation. For recent studies on estimation methods for static and time-varying spatial models, we refer to Lee and Yu (2010) and Blasques *et al.* (2016b), respectively.

2.2 Data and Descriptives

We use daily prices of 38 stock market indices similar to Asgharian *et al.* (2013) over the sample period from January 1995 to July 2020, leaving us with 6661 observations per time series.[1] Price data are sourced from Thomson Reuters Eikon. Continuously compounded returns are calculated as the log-price changes between consecutive trading days.

Table A.1 offers descriptive statistics for all data used in the empirical analysis. As indicated by the high levels of kurtosis, most time series significantly depart from normality. We therefore base our analysis on a multivariate t-distribution for the innovations in Equation 4. Results based on a multivariate normal distribution are available upon request.

To set up the weighting matrices, we source geographical distances between capital cities from the CEPII GeoDist database. Bilateral trade data and country-specific gross domestic products (GDP) are obtained from the OECD and ITC trademap database.

2.3 Spatial Weighting Matrix

The weighting matrix within a standard spatial model is defined by geographical distance. While a few decades ago, geographical proximity could have been regarded as a measure for economic proximity, the increase in mobility might have lead to a divergence between spatial and economic distances.

A variety of papers discusses the issue of measuring economic or financial integration in relation to geographical closeness. Portes and Rey (2005) use a gravity model to analyze the determinants of international trade in equities and show that investment

[1] In more detail, we consider the leading market indices of 6 American countries (Argentina, Brazil, Canada, Chile, Mexico, United States), 20 European countries (Austria, Belgium, Switzerland, Czech Republic, Germany, Denmark, Spain, Finland, France, United Kingdom, Greece, Hungary, Ireland, Israel, Netherlands, Norway, Poland, Portugal, Sweden, Turkey), and 12 Asian-Pacific countries (Australia, China, Hongkong, Indonesia, India, Japan, Korea, Malaysia, Philippines, Singapore, Thailand, Taiwan) in our empirical analysis.

in assets is cut in half, when the physical distance is doubled. This result is puzzling as investors could use more distant economies to diversify their risk. More specifically, they argue that information asymmetries between investors of different countries produce borders for international financial trade. Aviat and Coeurdacier (2007) build on the results of Portes and Rey (2005) and show that there is a clear relationship between bilateral trade and bilateral asset holdings between countries. More specifically, controlling for trade in the gravity model, the impact of physical distance is drastically reduced. Therefore, trade substitutes geological distance, while also accounting for financial interdependence. The idea of trade as a financial distance measure is also profound in the spatial statistics literature. For instance, Forbes and Chinn (2004) refer to trade linkages as the predominant determinant of market integration. On a similar note, Wälti (2011) find cross-border trade to foster market integration through both, the demand and the supply side. On the demand side, increases in aggregate demand for goods in one country lead to the export growth of its trading partners and, consequently, synchronization of business cycles. On the supply side, however, the impact is ambiguous and depends on whether trade occurs within (synchronization of business cycles) or across (de-synchronization of business cycles) industries.

Based on the discussion above, we use two definitions for the weighting matrix in Equation 9, which we refer to as geographic (W^{geo}) and economic weights (W^{eco}). The geographic weights are based on the geographic distance, while the economic weights are based on a bilateral trade measure. Following Asgharian *et al.* (2013), we obtain the GDP-standardized measure of bilateral trade between country i and country j as:

$$F_{ij,t} = 2 - \left(\frac{EXP_{ij,t}}{GDP_{i,t}} + \frac{EXP_{ji,t}}{GDP_{j,t}} \right), \tag{9}$$

where $EXP_{ij,t}$ is the nominal export value from country j to country i measured in US-

Dollars, and $GDP_{i,t}$ denotes the GDP of country i at time t. Besides, $EXP_{ji,t}$ denotes the nominal export value from country i to country j and $GDP_{j,t}$ refers to the GDP of country j. The measure of bilateral trade is obtained in quarterly frequency and employed with a lag of one period.

Figure 1 highlights that geographic and economic distance measures diverge considerably for some countries. While geographic distances cluster with respect to continents, economic distances are influenced by monetary, trade or regulatory unions. These observations highlight the importance of choosing adequate weighting matrices within spatial models.

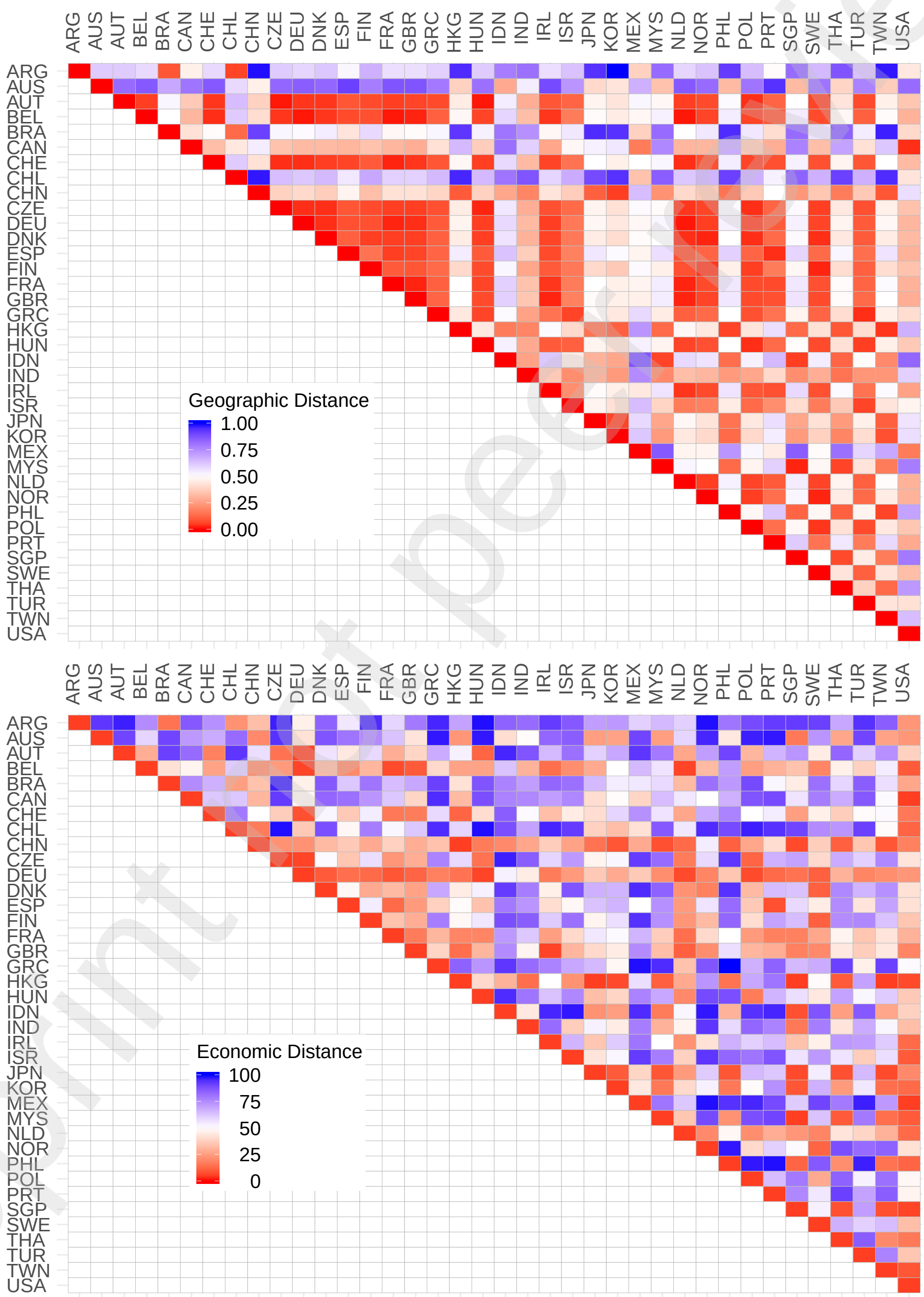
Denote the distances between units i and j for all $i = 1, \dots, n$ and $j = 1, \dots, n$ at time t in an $(n \times n)$ distance matrix D. The neighbourhood structure for any weight matrix W is obtained from D as:

$$W_{ij} = \begin{cases} 1 & \text{if } \mathrm{D}_{ij} \leq q_{10} \\ 0.75 & \text{if } \mathrm{q}_{10} < D_{ij} \leq q_{40} \\ 0.50 & \text{if } \mathrm{q}_{40} < D_{ij} \leq q_{60} \\ 0.25 & \text{if } \mathrm{q}_{60} < D_{ij} \leq q_{90} \\ 0 & \text{if } \mathrm{q}_{90} < D_{ij}, \end{cases} \tag{10}$$

where q_{10}, q_{40}, q_{60}, and q_{90} refer to the 10%, 40%, 60%, and 90% quantile of the empirical distribution of D_{ij}, respectively. Note, that the $i-$th row in W comprises all spatial weights impacting unit i. Thus, we set the main diagonal to zero and, in line with prior literature, row-standardize W such that for each i, $\sum_j W_{ij} = 1$ (e.g., Ord, 1975; Anselin, 1982). This results in an asymmetric spatial weight matrix W. Finally, the asymmetric spatial lag may be interpreted as a (geographically or economically) weighted average of all neighbours.

Figure 1: The relation between geographic- and economic distances

This figure depicts the relation between geographic- and economic distances for the entire cross-section of market indices. Geographic distances (upper panel) are scaled by the maximum geographic distance in our sample to be restricted within 0 and 100. Economic distances between countries (lower panel) are displayed by affiliation to the empirical distribution of mean economic distances over the full sample period. Both scales range from 0 (red, low distance) to 100 (blue, high distance).



3 Global equity market co-movement

3.1 Time-varying co-movement across equity markets

We estimate the SAR using the full set of sample countries (global) as well as subsets referring to the European, American, and Asian-Pacific regions. Figure 2 illustrates the global dynamic spatial dependence parameter based on the geographic (upper panel) and economic distance measure (lower panel).[2] Irrespective of the weighting scheme, we observe an increase in co-movement until the late 1990s. Afterwards the graph reveals constant or even decreasing co-movement in the early 2000s which then transitions into another upward trend until around 2010. The weighting scheme does not seem to have a decisive impact on the dynamics of the spatial dependence parameter. However, as obvious from the widening confidence bounds during crisis periods, volatility increases during crisis periods. As noted by Forbes and Rigobon (2002) this observed heteroscedasticity might bias contagion tests and it is therefore crucial to account for this pattern.

Figure 3 displays the time-varying co-movement parameters for American, European, and Asian-Pacific countries. We observed a higher level of spatial dependence for European countries compared to the American and Asian-Pacific countries. This result is hardly surprising, considering the geographic, economic, and political proximity of the European countries compared to the two other groups. The American countries tend to exhibit the lowest spatial dependence parameters, which jumps upwards during the first wave of the Covid-19 pandemic in 2020. Equity markets in the American and Asian-Pacific countries also seem to drive the trending patterns, which we observe for the global co-movement in Figure 2.

[2]Summary statistics on the estimation results can be found in table A.3.

Table 1 shows yearly averages over the estimated spatial parameters. The corresponding standard deviations can be found in Table A.2. These statistics confirm the indications from Figures 2 and 3 as they reveal the highest yearly average spatial dependence parameter for the European countries, followed by the Asian-Pacific, and the American countries. We further observe a slightly higher variation in the spatial dependence parameter on a yearly basis for Asian-Pacific countries compared to the European and American countries. Besides, we find an increase in co-movement across equity markets during some crisis periods. In 1998 global co-movement increases, potentially due to the Russian default. During the global financial crisis (GFC), co-movement increases as well, although only slightly, which might be due to the already very high level of co-movement. During 2016 another rise in global dependence occurs, which matches the timing of the Chinese stock market turbulence as well as the Brexit vote in the UK. The highest co-movement, however, is observed in the early months of 2020 when the Covid-19 disease starts to spread.

Figure 2: Time-varying spatial dependence between all markets based on dynamic volatilities: January 1995 to July 2020

This figure depicts mean estimates of the daily spatial dependence for all market indices over the sample period from January 1995 to July 2020. We provide time-varying estimates based on student-t distributed disturbances and time-varying volatility estimates. The red and blue line depict the 2.5% and 97.5% quantile of the distribution, respectively.

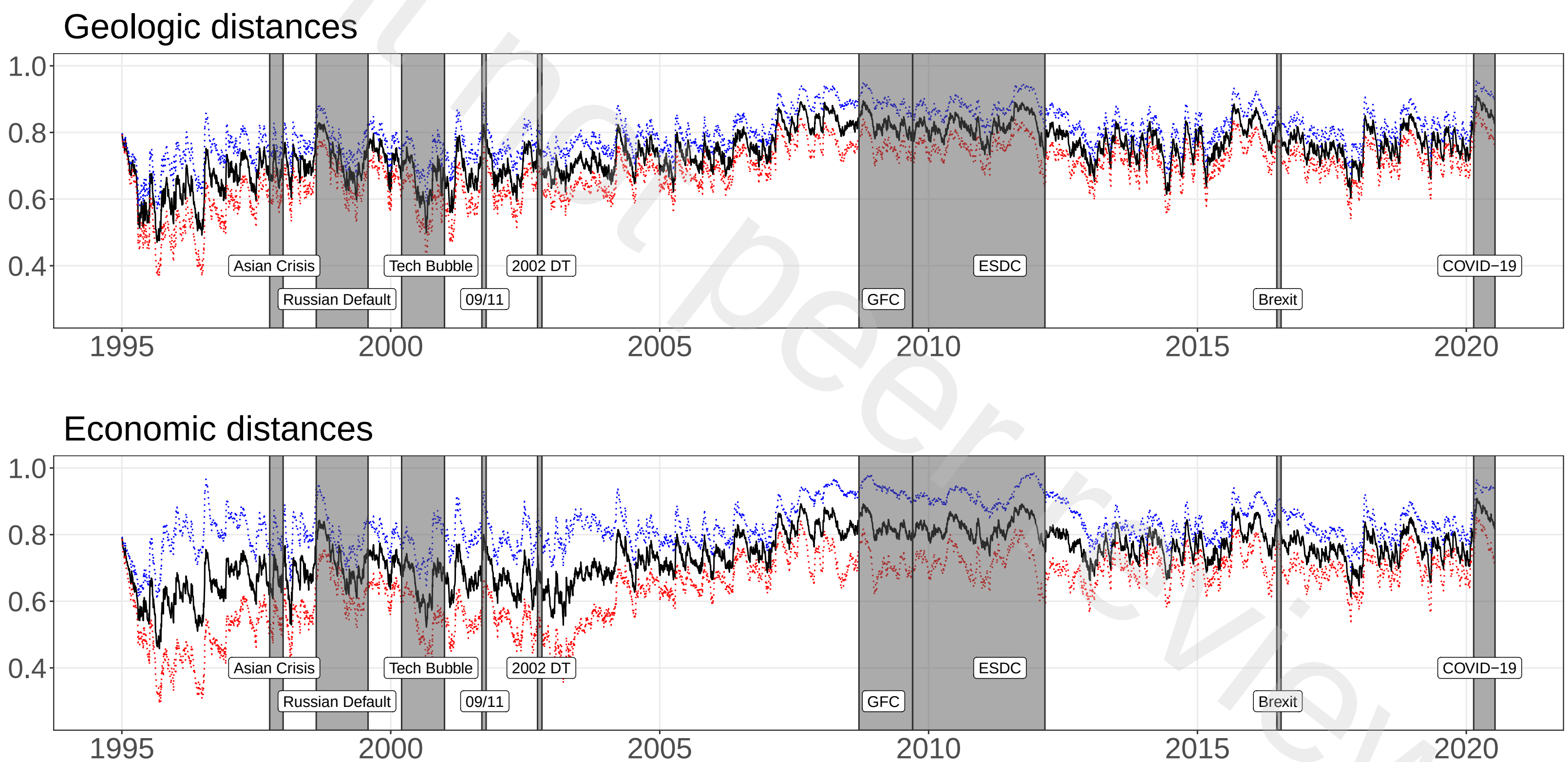


Figure 3: Time-varying spatial dependence within three market sets based on dynamic volatilities: January 1995 to July 2020

This figure depicts mean estimates of the daily spatial dependence for three market sets over the sample period from January 1995 to July 2020. The sets comprise European countries (dotted lines), American countries (dashed lines), and Asian-Pacific countries (solid lines). For each set of constituents, we provide time-varying estimates based on student-t distributed disturbances and time-varying volatility estimates.

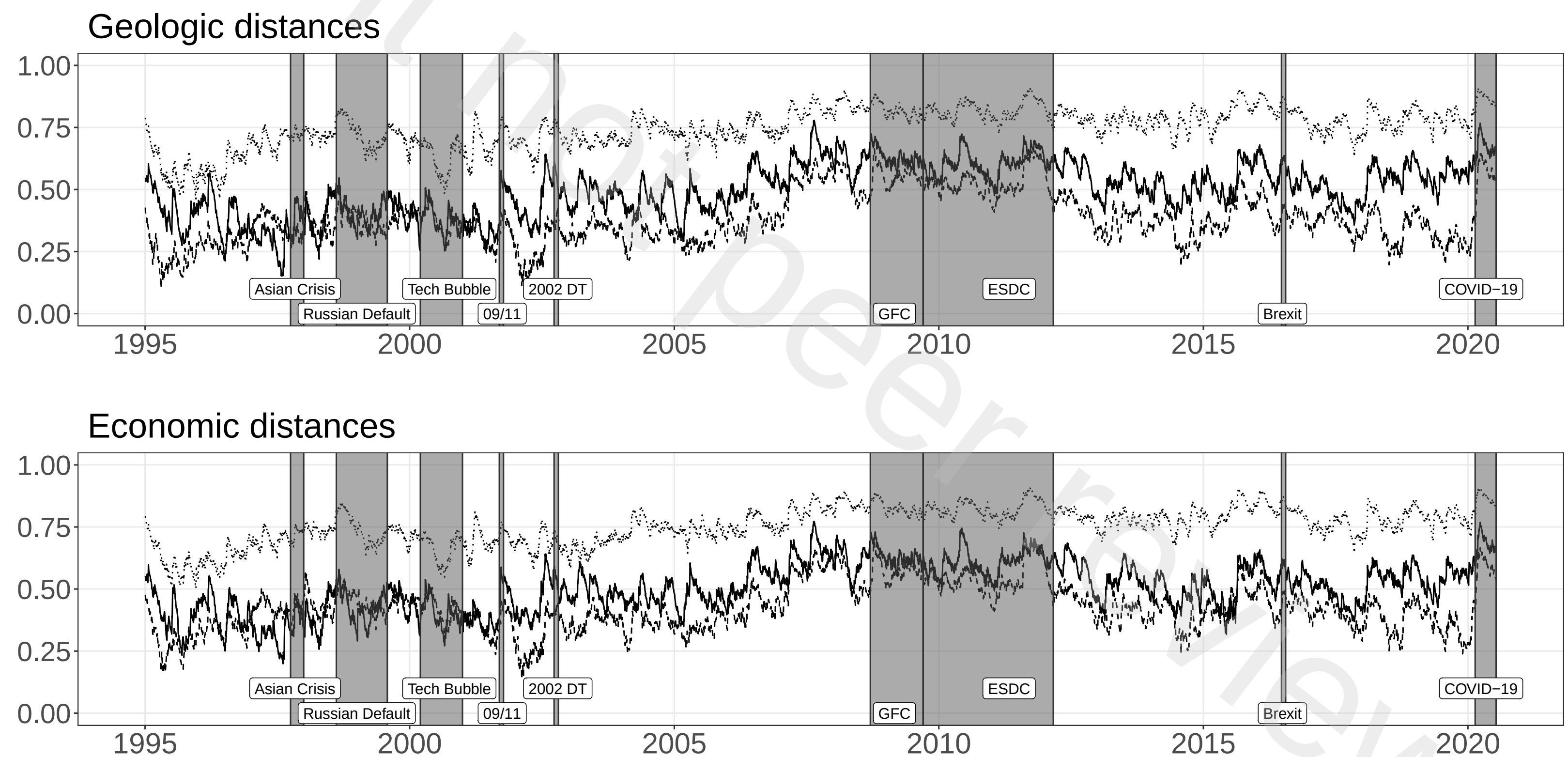


Table 1: Average yearly co-movement

This table provides the arithmetic mean of the time-varying spatial dependence parameter based on both the geographic and economic distance measure. In each case, we report values for the global and the regional samples from 1995 to 2020.

	(a) Geographic distance				**(b) Economic distance**			
	Global	**American**	**Asian-Pacific**	**European**	**Global**	**American**	**Asian-Pacific**	**European**
1995	0.6153	0.2314	0.5902	0.4192	0.6069	0.2899	0.6147	0.4015
1996	0.6325	0.2826	0.5984	0.3975	0.6321	0.3470	0.6257	0.4100
1997	0.6881	0.3578	0.7028	0.3184	0.6822	0.4252	0.7034	0.3307
1998	0.7346	0.3858	0.7459	0.3986	0.7203	0.4605	0.7629	0.4186
1999	0.7077	0.3661	0.6979	0.4091	0.7013	0.4310	0.7101	0.4348
2000	0.6679	0.3806	0.6435	0.3791	0.6694	0.4288	0.6720	0.4043
2001	0.6875	0.3277	0.6890	0.3872	0.6810	0.3636	0.6977	0.4048
2002	0.6922	0.2629	0.7030	0.4543	0.6499	0.3000	0.6806	0.4627
2003	0.6950	0.3405	0.6978	0.4844	0.6581	0.3733	0.6741	0.4935
2004	0.7312	0.3324	0.7475	0.4620	0.7234	0.3715	0.7547	0.4728
2005	0.7176	0.3012	0.7182	0.4335	0.7211	0.3492	0.7324	0.4606
2006	0.7481	0.3756	0.7444	0.5382	0.7520	0.4328	0.7586	0.5476
2007	0.8215	0.5305	0.8095	0.6235	0.8137	0.5649	0.8142	0.6277
2008	0.8418	0.5390	0.8432	0.6305	0.8376	0.5584	0.8476	0.6303
2009	0.8144	0.5389	0.8087	0.5990	0.8145	0.5595	0.8168	0.6024
2010	0.8123	0.5133	0.8159	0.6045	0.8162	0.5282	0.8211	0.6043
2011	0.8310	0.5457	0.8271	0.6211	0.8301	0.5619	0.8300	0.6276
2012	0.7740	0.4444	0.7940	0.5895	0.7794	0.4749	0.7986	0.5899
2013	0.7531	0.3633	0.7682	0.5154	0.7547	0.4003	0.7725	0.5274
2014	0.7466	0.3294	0.7712	0.4847	0.7613	0.3756	0.7760	0.4857
2015	0.7813	0.4119	0.8108	0.5403	0.7765	0.4558	0.8090	0.5004
2016	0.7944	0.4061	0.8270	0.5641	0.7963	0.4502	0.8282	0.5540
2017	0.7274	0.3633	0.7376	0.4671	0.7287	0.3932	0.7461	0.4674
2018	0.7760	0.3422	0.7836	0.5602	0.7660	0.3830	0.7828	0.5505
2019	0.7691	0.3145	0.7862	0.5571	0.7605	0.3580	0.7894	0.5394
2020	0.8422	0.5029	0.8419	0.6547	0.8357	0.5297	0.8420	0.6587

3.2 Contagion during recent major crises

We now turn to specific crisis periods and analyze the potential occurrence of contagion. Based on the definition of contagion by Forbes and Rigobon (2002), we examine whether there is evidence for contagion on a global or regional levels. For that purpose, we estimate confidence bounds based on the Delta-method in-sample confidence bands for time-varying parameters introduced by Blasques *et al.* (2016a). Since our model also allows for time-varying variances, these bounds also account for potentially increased volatility during crisis periods. More particularly, we examine potential contagion effects in the first 30 days of the Asian crisis, the 09/11 terrorist attacks, the 2002 downturn, the GFC, Brexit and the outbreak of Covid-19.

Adapting the definition by Forbes and Rigobon (2002), we define contagion as an increase in co-movement during a crisis period or in the direct aftermath of a crisis event. Consequently, we assess, whether the global an regional time-varying spatial dependence parameters increases significantly during the considered crisis periods. A significant increase is defined as non-overlapping of at least 7 continuous days of the crisis/post-event lower ($c_{t_c}^{LOW}$) confidence bounds and the arithmetic mean of the pre-event upper ($c_{t_p}^{UP}$) confidence bounds:

$$\begin{aligned} \frac{1}{N}\sum_{p=n}^{n+N} c_{t_p}^{UP} \ < \ c_{t_c}^{LOW} \qquad & t_c \in \{t_m, t_{m+1}, .., t_{m+M}\} \\ & \text{with} \quad t_{n+N} < t_m \qquad . \end{aligned} \tag{11}$$

To determine the crisis periods (t_c), we rely on previous literature, while the pre-crisis period is determined by the 30 days preceding the start of the crisis. We consider the 0.025 and 0.975 quantiles for the lower and upper bound, respectively. The results are given in Table 2.

With respect to the Asian crisis, Figures 2 and 3 reveal an increase in the co-movement among global and regional equity markets. However, we cannot detect a significant increase corresponding to non-overlapping pre-crisis upper and crisis lower confidence bounds as indicated in Table 2. Since our approach accounts for dynamic volatility, the co-movement parameters visible from Figure 3 are quite volatile, rendering a detection of contagion as defined in Equation 11 rather difficult. This is particularly noticeable when dealing with a rather long crisis period as is the case for the Asian crisis. Consequently, our results do not support those by Baig (1999), Caporale *et al.* (2005) and Chiang *et al.* (2007), who find evidence for regional market contagion over the Asian crisis period.

We also observe an increase in co-movement during the Russian default. In mid-1998, the Russian government defaults on domestic debt, which devalues the ruble and declares a moratorium on payment to foreign creditors. These developments triggered the collapse of the hedge fund LTCM, leading to turbulence within financial markets across various regions. According to Kenourgios *et al.* (2011b), the co-movement between Russian and other countries' equity markets increased significantly throughout the Russian default. Our results support theirs in so far that we do observe a rise co-movement. However, we may not detect contagion. Moreover, global equity market co-movement sharply increases from a monthly average of 0.68 in July 1998 to 0.79 in August and up to 0.83 in September 1998, reverting to pre-crisis levels in November 1998.

Turning to the tech bubble burst in the mid of March 2000 (e.g. Griffin *et al.*, 2011; Ljungqvist and Wilhelm Jr., 2003), we do not detect financial contagion. Surprisingly, we also do not detect global contagion for GFC, but rather non-significant increases in co-movement for global and regional samples, following the collapse of Lehman Brothers in September 2008, which is commonly considered as the starting date (e.g. Mollah *et al.*, 2016).

These findings are in line with those of Bekaert *et al.* (2014), who study contagion among equity markets during the GFC and find that there is weak evidence for global contagion.

The ongoing high co-movement after the end of the financial crisis of 2009 is likely due to the start of the European sovereign debt crisis. IC Pragidis *et al.* (2015) analyse contagion effects from 10-year government bond yields arising from the Greek debt crisis to other European countries, denoting a timeline of European debt crisis events and states the beginning in mid-2009 ranging to 2013. Moreover, Beirne and Fratzscher (2013) denote an increase of sovereign risk from 2008 to 2011, which ultimately results in the European Sovereign debt crisis 2010. Accordingly, fundamental contagion is observed during that time. Finally, Broto and Perez-Quiros (2015) define the beginning of the sovereign debt crisis in October 2010, going at least until March 2012 when Greece was hold accountable of its debt. Moreover, they find significant contagion throughout the sovereign debt crisis. Another rise in global and regional co-movement is observed in late 2015 to mid-2016, which might be attributed to the U.K.'s uncertainty leaving the European Union culminating in the Referendum in June 2016 and the decision for the Brexit.

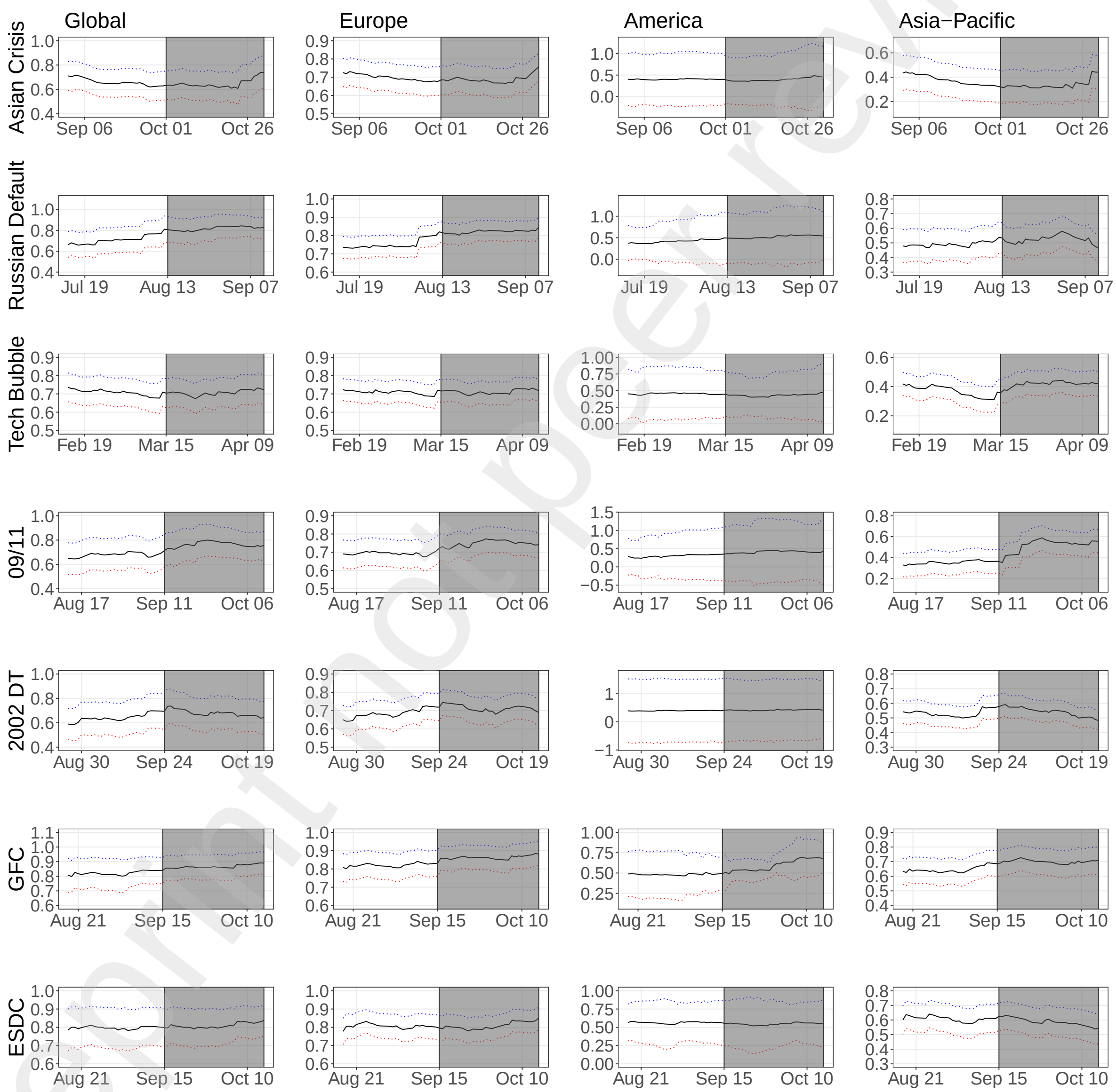
With respect to the pandemic induced crisis period starting in February 2020, we find evidence for global contagion. Overall, the economic downturn induced by the Covid-19 pandemic constitutes the only event for which we find a significant increase in co-movement among international equity markets despite the generally high level of co-movement brought about by increasing market integration within the last two decades. Overall, our findings support those by Forbes and Rigobon (2002). When accounting for heteroscedasticity in terms of increasing variance during crisis periods, we observe *no contagion, only interdependence*.

Table 2: Overview on selected events between 1997 and 2020
This table provides an overview on the selected event periods between January 1995 and July 2020. Besides, we tabulate the mean of the upper confidence bound for each pre-event period as well as the minimum of the lower confidence bound in contagion periods. Finally, the table highlights the presence of regional or global contagion for each of the considered events.

	Event period	mean($c_{t_p}^{UP}$)	**Contagion period**	min($c_{t_c}^{LOW}$)
Global				
Asian crisis	1997-10-01/1997-12-31	0.7774	No contagion	
Russian default	1998-08-13/1999-07-31	0.8339	No contagion	
Tech bubble	2000-03-15/2000-12-31	0.7878	No contagion	
09/11	2001-09-11/2001-10-10	0.8089	No contagion	
2002 DT	2002-09-24/2002-10-24	0.7745	No contagion	
GFC	2008-09-15/2009-09-14	0.9238	No contagion	
ESDC	2009-09-15/2012-03-01	0.9068	No contagion	
Brexit	2016-06-23/2016-07-22	0.8732	No contagion	
Covid-19	2020-02-20/2020-07-14	0.8264	2020-03-10/2020-04-08	0.8316
America				
Asian crisis	1997-10-01/1997-12-31	1.0176	No contagion	
Russian default	1998-08-13/1999-07-31	0.9052	No contagion	
Tech bubble	2000-03-15/2000-12-31	0.8355	No contagion	
09/11	2001-09-11/2001-10-10	0.9097	No contagion	
2002 DT	2002-09-24/2002-10-24	1.5249	No contagion	
GFC	2008-09-15/2009-09-14	0.7548	No contagion	
ESDC	2009-09-15/2012-03-01	0.8518	No contagion	
Brexit	2016-06-23/2016-07-22	0.6055	No contagion	
Covid-19	2020-02-20/2020-03-20	1.3513	No contagion	
Europe				
Asian crisis	1997-10-01/1997-12-31	0.7769	No contagion	
Russian default	1998-08-13/1999-07-31	0.8121	No contagion	
Tech bubble	2000-03-15/2000-12-31	0.7712	No contagion	
09/11	2001-09-11/2001-10-10	0.7686	No contagion	
2002 DT	2002-09-24/2002-10-24	0.7568	No contagion	
GFC	2008-09-15/2009-09-14	0.8926	No contagion	
ESDC	2009-09-15/2012-03-01	0.8733	No contagion	
Brexit	2016-06-23/2016-07-22	0.8892	No contagion	
Covid-19	2020-02-20/2020-07-14	0.8418	2020-03-13/2020-03-20	0.8440
			2020-03-25/2020-04-02	0.8442
Asia-Pacific				
Asian crisis	1997-10-01/1997-12-31	0.5158	No contagion	
Russian default	1998-08-13/1999-07-31	0.6009	No contagion	
Tech bubble	2000-03-15/2000-12-31	0.4561	No contagion	
09/11	2001-09-11/2001-10-10	0.4649	No contagion	
2002 DT	2002-09-24/2002-10-24	0.6066	No contagion	
GFC	2008-09-15/2009-09-14	0.7329	No contagion	
ESDC	2009-09-15/2012-03-01	0.7077	No contagion	
Brexit	2016-06-23/2016-07-22	0.6140	No contagion	
Covid-19	2020-02-20/2020-07-14	0.6600	2020-03-23/2020-04-03	0.6670

Figure 4: Financial contagion: Selected events between 1997 and 2020

This figure depicts mean estimates (black lines) and confidence intervals (red and blue dotted lines) of the time-varying spatial dependence between one global and three regional sets of stock markets over 60-day-windows around selected economic or political events between 1997 and 2020. More particularly, we demonstrate potential contagion effects in the first 30 days of the Asian crisis, the 09/11 terrorist attacks, the 2002 downturn, the GFC, Brexit and the outbreak of Covid-19.



4 Concluding remarks

We empirically examine the co-movement among equity markets around the world. Based on a time-varying spatial model, we estimate the global co-movement of 38 countries as well as for subsamples of American, Asian-Pacific and European countries from 1995 to 2020. Our approach not only is able to capture the time-variation within the co-movement, but also accounts for increasing variances during event or crisis periods. Using confidence intervals, we are able to detect contagion, measured by a significant increase in co-movement during an event. First, our results indicate that contagion on a global level occurred only once within the last 25 years, namely during the first wave of the Covid-19 pandemic in the first quarter of 2020. Second, our results highlight that equity market co-movement increased during the last two decades, in particular among European countries, while generally exhibiting a high level of fluctuation. In contrast, standard approaches such as correlation analysis neglect the time-varying behavior of co-movement. For this reason our results provide a more general view of co-movement beyond pairwise correlation and highlight the high level of global equity market integration. Last, and in support of the findings by Forbes and Rigobon (2002), market dynamics often considered as contagion can be rather described as interdependence when accounting for time-varying variances.

5 Appendix

Table A.1: Descriptive statistics

This table summarizes the data used in the empirical analysis. The data set covers the period from January 1995 to April 2020. The table offers descriptive statistics of the (log-) return series (r) for each equity index. In more detail, we report sample mean, median, standard deviation, minimum, maximum, skewness, kurtosis, and lag-1 autocorrelation of the time series.

	Mean	Median	St. Dev.	Min.	Max.	Skewness	Kurtosis	Autocorr.
r_{ARG}	0.0007	0.0001	0.0229	-0.4769	0.1612	-1.6174	32.2991	0.0339
r_{AUS}	0.0002	0.0002	0.0098	-0.1020	0.0677	-0.7074	8.9089	-0.0487
r_{AUT}	0.0001	0.0000	0.0136	-0.1467	0.1202	-0.6161	10.2420	0.0650
r_{BEL}	0.0001	0.0002	0.0121	-0.1533	0.0933	-0.4255	9.8760	0.0719
r_{BRA}	0.0005	0.0000	0.0206	-0.1721	0.2883	0.2750	14.8819	-0.0016
r_{CAN}	0.0002	0.0005	0.0107	-0.1318	0.1129	-0.9794	17.4759	-0.0282
r_{CHE}	0.0002	0.0003	0.0115	-0.1013	0.1079	-0.2964	7.3524	0.0283
r_{CHL}	0.0002	0.0000	0.0084	-0.1384	0.0906	-0.9685	26.3995	0.1645
r_{CHN}	0.0003	0.0000	0.0168	-0.1843	0.2785	0.2285	18.6131	0.0173
r_{CZE}	0.0001	0.0000	0.0128	-0.1619	0.1236	-0.5162	13.4201	0.0869
r_{DEU}	0.0003	0.0006	0.0144	-0.1305	0.1080	-0.2389	5.8086	-0.0089
r_{DNK}	0.0002	0.0002	0.0168	-0.1740	0.1456	-0.4230	8.1888	0.0170
r_{ESP}	0.0004	0.0003	0.0119	-0.1172	0.0950	-0.3446	5.6899	0.0379
r_{FIN}	0.0001	0.0004	0.0143	-0.1515	0.1348	-0.3499	7.7566	0.0129
r_{FRA}	0.0001	0.0002	0.0140	-0.1310	0.1059	-0.2195	6.0726	-0.0153
r_{GBR}	0.0001	0.0001	0.0114	-0.1151	0.0938	-0.3315	7.9563	-0.0186
r_{GRC}	-0.0001	0.0000	0.0183	-0.1771	0.1343	-0.4217	7.0551	0.0923
r_{HKG}	0.0002	0.0000	0.0154	-0.1473	0.1725	0.0510	10.8779	-0.0069
r_{HUN}	0.0005	0.0002	0.0161	-0.1803	0.1362	-0.6186	12.0170	0.0497
r_{IDN}	0.0004	0.0001	0.0145	-0.1273	0.1313	-0.1917	9.4629	0.1388
r_{IND}	0.0003	0.0000	0.0147	-0.1410	0.1599	-0.2244	8.1023	0.0485
r_{IRL}	0.0002	0.0003	0.0129	-0.1396	0.0973	-0.7474	9.5372	0.0535
r_{ISR}	0.0003	0.0000	0.0120	-0.1038	0.0978	-0.3406	7.0047	0.0305
r_{JPN}	0.0000	0.0000	0.0145	-0.1211	0.1323	-0.2893	6.0806	-0.0319
r_{KOR}	0.0001	0.0000	0.0164	-0.1280	0.1128	-0.2142	6.5093	0.0488
r_{MEX}	0.0004	0.0001	0.0141	-0.1431	0.1215	0.0336	7.8200	0.0909
r_{MYS}	0.0001	0.0000	0.0120	-0.2415	0.2082	0.5002	66.7282	0.0537
r_{NLD}	0.0002	0.0004	0.0136	-0.1138	0.1003	-0.2458	7.0884	0.0047
r_{NOR}	0.0002	0.0002	0.0143	-0.1128	0.1102	-0.5792	7.2698	-0.0046
r_{PHL}	0.0001	0.0000	0.0137	-0.1432	0.1618	-0.1027	13.1552	0.1204
r_{POL}	0.0003	0.0000	0.0139	-0.1353	0.0789	-0.3861	5.6660	0.0976
r_{PRT}	0.0000	0.0000	0.0116	-0.1038	0.1020	-0.4825	8.1315	0.0973
r_{SGP}	0.0000	0.0000	0.0123	-0.0983	0.1097	-0.0380	7.2235	0.0470
r_{SWE}	0.0003	0.0001	0.0143	-0.1117	0.1102	-0.0486	4.7122	-0.0168
r_{THA}	0.0000	0.0000	0.0146	-0.1606	0.1135	-0.0763	9.5571	0.0623
r_{TUR}	0.0009	0.0002	0.0226	-0.1998	0.1777	-0.0688	6.9176	0.0203
r_{TWN}	0.0001	0.0000	0.0134	-0.0994	0.0852	-0.2320	3.9193	0.0290
r_{USA}	0.0003	0.0003	0.0119	-0.1277	0.1096	-0.4163	11.3581	-0.0979

Table A.2: Average yearly standard deviation of the co-movement parameter
This table provides the yearly standard deviation of the time-varying spatial dependence parameter based on both the geographic and economic distance measure. In each case, we report values for the global and the regional samples from 1995 to 2020.

	(a) Geographic distance				(b) Economic distance			
	Global	**American**	**Asian Pacific**	**European**	**Global**	**American**	**Asian-Pacific**	**European**
1995	0.0797	0.0602	0.0732	0.0816	0.0727	0.0705	0.0685	0.0899
1996	0.0601	0.0339	0.0568	0.0798	0.0555	0.0362	0.0394	0.0698
1997	0.0351	0.0431	0.0331	0.0715	0.0362	0.0312	0.0310	0.0587
1998	0.0511	0.0518	0.0402	0.0747	0.0691	0.0498	0.0398	0.0741
1999	0.0472	0.0354	0.0404	0.0470	0.0403	0.0325	0.0343	0.0575
2000	0.0579	0.0475	0.0661	0.0564	0.0566	0.0461	0.0552	0.0555
2001	0.0612	0.0489	0.0640	0.0826	0.0499	0.0519	0.0465	0.0702
2002	0.0417	0.0806	0.0547	0.0902	0.0400	0.0845	0.0411	0.0797
2003	0.0237	0.0338	0.0227	0.0472	0.0442	0.0372	0.0346	0.0467
2004	0.0423	0.0523	0.0391	0.0536	0.0405	0.0516	0.0344	0.0404
2005	0.0343	0.0352	0.0318	0.0583	0.0306	0.0343	0.0276	0.0502
2006	0.0332	0.0430	0.0351	0.0584	0.0337	0.0465	0.0336	0.0610
2007	0.0418	0.0622	0.0410	0.0699	0.0432	0.0632	0.0392	0.0657
2008	0.0280	0.0620	0.0262	0.0566	0.0305	0.0661	0.0241	0.0605
2009	0.0168	0.0265	0.0209	0.0319	0.0148	0.0266	0.0179	0.0314
2010	0.0249	0.0331	0.0310	0.0503	0.0255	0.0368	0.0278	0.0554
2011	0.0406	0.0744	0.0449	0.0551	0.0393	0.0850	0.0438	0.0520
2012	0.0338	0.0540	0.0275	0.0507	0.0333	0.0487	0.0239	0.0562
2013	0.0381	0.0469	0.0319	0.0443	0.0350	0.0426	0.0298	0.0567
2014	0.0529	0.0614	0.0438	0.0495	0.0406	0.0624	0.0402	0.0516
2015	0.0540	0.0690	0.0516	0.0686	0.0487	0.0572	0.0456	0.0871
2016	0.0337	0.0502	0.0327	0.0511	0.0335	0.0496	0.0317	0.0449
2017	0.0358	0.0462	0.0361	0.0458	0.0350	0.0417	0.0341	0.0472
2018	0.0480	0.0669	0.0446	0.0546	0.0454	0.0641	0.0437	0.0504
2019	0.0353	0.0511	0.0338	0.0390	0.0353	0.0578	0.0314	0.0476
2020	0.0521	0.1270	0.0532	0.0532	0.0559	0.1189	0.0504	0.0579

Table A.3: Coefficient estimates of the SAR: January 1995 to July 2020

This table provides results for the SAR models with time-varying co-movement and volatility based on the sample period from January 1995 to July 2020. For each combination of constituents, we report coefficients based on geographic and economic spatial weight matrices and student-t distributed error terms. Besides, we provide the Log-Likehood (LL), the Akaike information criterion (AIC), and the Bayesian information criterion (BIC). ***, **, and * denote statistical significance at the 1%, 5%, and 10% level, respectively.

	Global		Europe		America		Asia-Pacific	
	W^{geo}	W^{eco}	W^{geo}	W^{eco}	W^{geo}	W^{eco}	W^{geo}	W^{eco}
β	0.0001***	0.0002***	0.0001***	0.0001***	0.0004***	0.0003***	0.0002***	0.0003***
ω	0.0102***	0.0088**	0.0073***	0.0071***	0.0021***	0.0029	0.0036***	0.0042***
A	0.0083***	0.0079***	0.0072***	0.0073***	0.0046***	0.0055	0.0065***	0.0067***
B	0.9897***	0.9910***	0.9927***	0.9931***	0.9951***	0.9939***	0.9940***	0.9930***
ν	8.5904***	8.1341***	7.8046***	7.7612***	6.9335***	7.1466***	8.7099***	8.7940***
ω^{σ}_{ARG}	-0.0410***	-0.0411***			-0.0794***	-0.0789***		
ω^{σ}_{AUS}	-0.0492***	-0.0494***					-0.0483***	-0.0489***
ω^{σ}_{AUT}	-0.0478***	-0.0480***	-0.0398***	-0.0407***				
ω^{σ}_{BEL}	-0.0508***	-0.0506***	-0.0427***	-0.0435***				
ω^{σ}_{BRA}	-0.0425***	-0.0425***			-0.0821***	-0.0817***		
ω^{σ}_{CAN}	-0.0502***	-0.0503***			-0.0983***	-0.0977***		
ω^{σ}_{CHE}	-0.0506***	-0.0505***	-0.0421***	-0.0432***				
ω^{σ}_{CHL}	-0.0502***	-0.0509***			-0.0982***	-0.0975**		
ω^{σ}_{CHN}	-0.0433***	-0.0435***					-0.0437***	-0.0439***
ω^{σ}_{CZE}	-0.0480***	-0.0481***	-0.0395***	-0.0403***				
ω^{σ}_{DEU}	-0.0489***	-0.0485***	-0.0412***	-0.0418***				
ω^{σ}_{DNK}	-0.0466***	-0.0469***	-0.0392***	-0.0402***				
ω^{σ}_{ESP}	-0.0478***	-0.0480***	-0.0397***	-0.0406***				
ω^{σ}_{FIN}	-0.0477***	-0.0480***	-0.0400***	-0.0414***				
ω^{σ}_{FRA}	-0.0493***	-0.0490***	-0.0417***	-0.0424***				
ω^{σ}_{GBR}	-0.0510***	-0.0508***	-0.0427***	-0.0436***				
ω^{σ}_{GRC}	-0.0438***	-0.0439***	-0.0362***	-0.0369***				
ω^{σ}_{HKG}	-0.0458***	-0.0458***					-0.0449***	-0.0451***
ω^{σ}_{HUN}	-0.0445***	-0.0447***	-0.0370***	-0.0377***				
ω^{σ}_{IDN}	-0.0461***	-0.0462***					-0.0459***	-0.0460***
ω^{σ}_{IND}	-0.0453***	-0.0455***					-0.0447***	-0.0448***
ω^{σ}_{IRL}	-0.0483***	-0.0484***	-0.0402***	-0.0411***				
ω^{σ}_{ISR}	-0.0473***	-0.0475***	-0.0377***	-0.0396***				
ω^{σ}_{JPN}	-0.0452***	-0.0453***					-0.0449***	-0.0451***
ω^{σ}_{KOR}	-0.0453***	-0.0455***					-0.0452***	-0.0454***
ω^{σ}_{MEX}	-0.0466***	-0.0467***			-0.0901***	-0.0897*		
ω^{σ}_{MYS}	-0.0493***	-0.0495***					-0.0499***	-0.0498***
ω^{σ}_{NLD}	-0.0508***	-0.0503***	-0.0429***	-0.0437***				
ω^{σ}_{NOR}	-0.0472***	-0.0474***	-0.0392***	-0.0400***				
ω^{σ}_{PHL}	-0.0449***	-0.0451***					-0.0448***	-0.0448***
ω^{σ}_{POL}	-0.0464***	-0.0465***	-0.0383***	-0.0391***				
ω^{σ}_{PRT}	-0.0490***	-0.0492***	-0.0405***	-0.0414***				
ω^{σ}_{SGP}	-0.0486***	-0.0485***					-0.0479***	-0.0481***
ω^{σ}_{SWE}	-0.0478***	-0.0482***	-0.0402***	-0.0413***				
ω^{σ}_{THA}	-0.0461***	-0.0463***					-0.0458***	-0.0459***
ω^{σ}_{TUR}	-0.0409***	-0.0411***	-0.0338***	-0.0344***				
ω^{σ}_{TWN}	-0.0459***	-0.0461***					-0.0459***	-0.0461***
ω^{σ}_{USA}	-0.0494***	-0.0493***			-0.0961***	-0.0952***		
A^{σ}	0.0844***	0.0847***	0.0837***	0.0847***	0.1457***	0.1438***	0.1142***	0.1130***
B^{σ}	0.9951***	0.9951***	0.9960***	0.9959***	0.9905***	0.9906***	0.9951***	0.9951***
LL	833026	829936	451604	451955	129621	129973	256784	256980
AIC	-1665962	-1659782	-903155	-903855	-259216	-259920	-513530	-513921
BIC	-1665656	-1659476	-902971	-903671	-259127	-259831	-513401	-513792

References

Anselin, L. (1982) A note on small sample properties of estimators in a first-order spartial autoregressive model, *Environment and Planning A*, **14**, 1023–1030.

Arnold, M., Stahlberg, S. and Wied, D. (2013) Modeling different kinds of spatial dependence in stock returns, *Empirical Economics*, **44**, 761–774.

Asgharian, H., Hess, W. and Lu, L. (2013) A spatial analysis of international stock market linkages, *Journal of Banking and Finance*, **37**, 4738–4754.

Aviat, A. and Coeurdacier, N. (2007) The geography of trade in goods and asset holdings, *Journal of International Economics*, **71**, 22 – 51.

Bae, K.-H., Karolyi, G. A. and Stulz, R. M. (2003) A new approach to measuring financial contagion, *The Review of Financial Studies*, **16**, 717–763.

Baig, I., T. Goldfajn (1999) Financial market contagion in the Asian crisis, *IMF Economic Review*, **46**, 167–195.

Beirne, J. and Fratzscher, M. (2013) The pricing of sovereign risk and contagion during the european sovereign debt crisis, *Journal of International Money and Finance*, **34**, 60–82.

Bekaert, G., Ehrmann, M., Fratscher, M. and Mehl, A. (2014) The Global Crisis and Equity Market Contagion, *The Journal of Finance*, **69**, 2597–2649.

Bekaert, G., Hodrick, R. J. and Zhang, X. (2009) International stock return comovements, *The Journal of Finance*, **64**, 2591–2626.

Blasques, F., Koopman, S. J., Lasak, K. and Lucas, A. (2016a) In-sample confidence bands and out-of-sample forecast bands for time-varying parameters in observation-driven models, *International Journal of Forecasting*, **32**, 875–887.

Blasques, F., Koopman, S. J., Lucas, A. and Schaumburg, J. (2016b) Spillover dynamics for systemic risk measurement using spatial financial time series models, *Journal of Econometrics*, **195**, 255–271.

Boubaker, S., Jouini, J. and Lahiani, A. (2016) Financial contagion between the US and selected developed and emerging countries: The case of the subprime crisis, *The Quarterly Review of Economics and Finance*, **61**, 14–28.

Broto, C. and Perez-Quiros, G. (2015) Disentangling contagion among sovereign cds spreads during the european debt crisis, *Journal of Empirical Finance*, **32**, 165–179.

Caporale, G. M., Cipollini, A. and Spagnolo, N. (2005) Testing for contagion: A conditional correlation analysis, *Journal of Empirical Finance*, **12**, 476–489.

Chiang, T. C., Jeon, B. N. and Li, H. (2007) Dynamic correlation analysis of financial contagion: Evidence from Asian markets, *Journal of International Money and Finance*, **26**, 1206–1228.

Cliff, A. D. and Ord, J. K. (1973) Spatial autocorrelation, London, Pion.

Cliff, A. D. and Ord, J. K. (1981) Spatial processes, models and applications, London, Pion.

Creal, D., Koopman, S. J. and Lucas, A. (2011) A dynamic multivariate heavy-tailed model for time-varying volatilities and correlations, *Journal of Business and Economic Statistics*, **29**, 552–563.

Creal, D., Koopman, S. J. and Lucas, A. (2013) Generalized autoregressive score models with applications, *Journal of Applied Econometrics*, **28**, 777–795.

Debreu, G. and Herstein, I. N. (1953) Nonnegative square matrices, *Econometrica*, **21**, 597–607.

Fernandez, V. (2011) Spatial linkages in international financial markets, *Quantitative Finance*, **11**, 237–245.

Forbes, K. (2012) The 'Big C': Identifying contagion, Working Paper 18465, National Bureau of Economic Research.

Forbes, K. J. and Chinn, M. D. (2004) A decomposition of global linkages in financial markets over time, *Review of Economics and Statistics*, **86**, 705–722.

Forbes, K. J. and Rigobon, R. (2002) No contagion, only interdependence: Measuring stock market comovements, *The Journal of Finance*, **57**, 2223–2261.

Griffin, H. J. H., John M., Shu, T. and Topaloglu, S. (2011) Who drove and burst the tech bubble?, *The Journal of Finance*, **66**, 1251–1290.

Harvey, A. and Luati, A. (2014) Filtering with heavy tails, *Journal of the American Statistical Association*, **109**, 1112–1122.

Harvey, A. C. (2013) *Dynamic models for volatility and heavy tails*, Cambridge University Press, in: Econometric Society Monographs.

IC Pragidis, I., Aielli, G., Chionis, D. and Schizas, P. (2015) Contagion effects during financial crisis: Evidence from the Greek sovereign bonds market, *Journal of Financial Stability*, **18**, 127–138.

Kelejian, H. H. and Prucha, I. R. (2010) Specification and estimation of spatial autoregressive models with autoregressive and heteroskedastic disturbances, *Journal of Econometrics*, **157**, 53–67.

Kenourgios, D., Samitas, A. and Paltalidis, N. (2011a) Financial crises and stock market contagion in a multivariate time-varying asymmetric framework, *Journal of International Financial Markets, Institutions and Money*, **21**, 92–106.

Kenourgios, D., Samitas, A. and Paltalidis, N. (2011b) Financial crises and stock market contagion in a multivariate time-varying asymmetric framework, *Journal of International Financial Markets, Institutions and Money*, **21**, 92–106.

Krugman, P. (1992a) A dynamic spatial model, NBER Working Papers 4219, National Bureau of Economic Research, Inc.

Krugman, P. (1992b) *Geography and trade*, vol. 1 of *MIT Press Books*, The MIT Press.

Lee, L. (2004) Asymptotic distributions for quasi-maximum likelihood estimators for spatial autoregressive models, *Econometrica*, **72**, 1899–1925.

Lee, L. and Yu, J. (2010) Some recent developments in spatial panel data models, *Regional Science and Urban Economics*, **40**, 255–271.

Lesage, J. P. and Pace, R. K. (2009) *Introduction to spatial econometrics*, CRC Press.

Ljungqvist, A. and Wilhelm Jr., W. J. (2003) Ipo pricing in the dot-com bubble, *The Journal of Finance*, **58**.

Londono, J. M. (2019) Bad bad contagion, *Journal of Banking and Finance*, **108**.

Longin, F. and Solnik, B. (2001) Extreme correlation of international equity markets, *The Journal of Finance*, **56**, 649–676.

Mollah, S., Quoreshi, S. and Zafirov, G. (2016) Equity market contagion during global financial and Eurozone crises: Evidence from a dynamic correlation analysis, *Journal of International Financial Markets, Institutions and Money*, **41**, 151–167.

Ord, J. K. (1975) Estimation methods for models of spatial interaction, *Journal of the American Statistical Association*, **70**, 120–126.

Portes, R. and Rey, H. (2005) The determinants of cross-border equity flows, *Journal of International Economics*, **65**, 269–296.

Qu, X. and Lee, L.-f. (2015) Estimating a spatial autoregressive model with an endogenous spatial weight matrix, *Journal of Econometrics*, **184**, 209–232.

Rodriguez, J. C. (2007) Measuring financial contagion: A copula approach, *Journal of Empirical Finance*, **14**, 401–423.

Wälti, S. (2011) Stock market synchronization and monetary integration, *Journal of International Money and Finance*, **30**, 96–110.